

Table 1 Optimal Solution Value and IBFS Solution Results from 7 Methods

No	Name	VAM	JHM	TOCM-MT	BCE	SSM	CSM	Optimal Solution
1	N01	5600	5600	5600	5600	5600	5600	5600
2	N02	28	28	28	28	28	28	28
3	N03	1104	1128	1127	1137	1134	1134	1102
4	N04	1580	1580	1580	1580	1625	1580	1580
5	N05	470	420	435	410	430	430	410
6	N06	8350	7750	7750	8350	7750	7750	7750
7	N07	1220	1170	1160	1170	1165	1160	1160
8	N09	114	114	111	111	111	111	111
9	N10	990	960	910	910	960	910	910
10	N11	2980	2500	2500	2460	2500	2460	2460
11	N12	425	439	439	439	425	439	425
12	N13	5125	4525	5225	4525	4525	4525	4525
13	N14	960	920	930	920	920	920	920
14	N15	156	156	156	156	156	156	156
15	N16	1590	1530	1530	1530	1510	1510	1510
16	N17	779	743	743	743	743	779	743
17	N18	1745	1650	1650	1695	1650	1650	1650
18	N21	4205	4429	4205	4429	4205	4429	4205
19	N22	12075	12075	12200	12700	12075	12075	12075
20	N23	23	23	23	23	23	23	23
21	N24	3520	3460	3460	3460	3460	3460	3460
22	N25	849	809	809	809	809	849	809
23	N26	490	490	508	490	490	490	490
24	N27	68804	64499	59356	64499	59356	59356	59356
25	N28	2690	2090	2090	2090	2090	2090	2090
26	N29	3857	3857	3857	3857	3857	3857	3857
27	N30	473	475	475	475	465	475	465
28	N31	1480	1480	1480	1480	1480	1480	1480
29	N32	1780	1780	1780	1780	1820	1780	1780
30	N33	1430	1310	1250	1310	1310	1310	1250
31	N34	510	510	510	510	510	510	510
32	N35	320	280	280	280	280	280	280
33	S1	18855	20925	18855	19035	20925	18855	18855
34	S2	153500	176500	153500	153500	181500	153500	153500
35	S3	1044180	1044180	1044180	1044180	1044180	1044180	1044180
36	S4	93351	95301	92431	94264	95301	95440	92212
37	S5	116020	116020	116020	116020	116020	116020	116020
38	Real 1	12928429	12928429	12965785	12928429	12928429	12928429	12928429
39	Real 2	13657053	13657053	13700907	13657053	13657053	13657053	13657053
40	Real 3	12296587	12296587	12322885	12296587	12296587	12296587	12296587
41	Real 4	14117586	14117586	14166940	14117586	14117586	14117586	14117586
42	Real 5	14017419	14017419	14074887	14017419	14017419	14017419	14017419

Table 2 The Improvement Percentage (%) of CSM over VAM, JHM, TOCM-MT, BCE, and SSM.

No	Name	VAM	JHM	TOCM-MT	BCE	SSM
1	N01	0.00	0.00	0.00	0.00	0.00
2	N02	0.00	0.00	0.00	0.00	0.00
3	N03	-2.72	-0.53	-0.62	0.26	0.00
4	N04	0.00	0.00	0.00	0.00	2.77
5	N05	8.51	-2.38	1.15	-4.88	0.00
6	N06	7.19	0.00	0.00	7.19	0.00
7	N07	4.92	0.85	0.00	0.85	0.43
8	N09	2.63	2.63	0.00	0.00	0.00
9	N10	8.08	5.21	0.00	0.00	5.21
10	N11	17.45	1.60	1.60	0.00	1.60
11	N12	-3.29	0.00	0.00	0.00	-3.29
12	N13	11.71	0.00	13.40	0.00	0.00
13	N14	4.17	0.00	1.08	0.00	0.00
14	N15	0.00	0.00	0.00	0.00	0.00
15	N16	5.03	1.31	1.31	1.31	0.00
16	N17	0.00	-4.85	-4.85	-4.85	-4.85
17	N18	5.44	0.00	0.00	2.65	0.00
18	N21	-5.33	0.00	-5.33	0.00	-5.33
19	N22	0.00	0.00	1.02	4.92	0.00
20	N23	0.00	0.00	0.00	0.00	0.00
21	N24	1.70	0.00	0.00	0.00	0.00
22	N25	0.00	-4.94	-4.94	-4.94	-4.94
23	N26	0.00	0.00	3.54	0.00	0.00
24	N27	13.73	7.97	0.00	7.97	0.00
25	N28	22.30	0.00	0.00	0.00	0.00
26	N29	0.00	0.00	0.00	0.00	0.00
27	N30	-0.42	0.00	0.00	0.00	-2.15
28	N31	0.00	0.00	0.00	0.00	0.00
29	N32	0.00	0.00	0.00	0.00	2.20
30	N33	8.39	0.00	-4.80	0.00	0.00
31	N34	0.00	0.00	0.00	0.00	0.00
32	N35	12.50	0.00	0.00	0.00	0.00
33	S1	0.00	9.89	0.00	0.95	9.89
34	S2	0.00	13.03	0.00	0.00	15.43
35	S3	0.00	0.00	0.00	0.00	0.00
36	S4	-2.24	-0.15	-3.26	-1.25	-0.15
37	S5	0.00	0.00	0.00	0.00	0.00
38	Real 1	0.00	0.00	0.29	0.00	0.00
39	Real 2	0.00	0.00	0.32	0.00	0.00
40	Real 3	0.00	0.00	0.21	0.00	0.00
41	Real 4	0.00	0.00	0.35	0.00	0.00
42	Real 5	0.00	0.00	0.41	0.00	0.00
	Average	2.85	0.71	0.02	0.24	0.40

Table 3 The Improvement Percentage (%) of CSM over VAM, JHM, TOCM-MT, BCE, and SSM.

NO	Method	Total of			Peform (%)		
		Positive	Negative	Zero	Positive	Negative	Zero
1	VAM	15	5	22	33%	11%	49%
2	JHM	9	5	29	20%	11%	64%
3	TOCM-MT	13	6	24	29%	13%	53%
4	BCE	9	4	30	20%	9%	67%
5	SSM	8	6	29	18%	13%	64%

Table 4 The deviation percentage (%) of VAM, JHM, TOCM-MT, BCE, SSM, and CSM from the optimal solution.

No	Name	VAM	JHM	TOCM-MT	BCE	SSM	CSM
1	N01	0.00	0.00	0.00	0.00	0.00	0.00
2	N02	0.00	0.00	0.00	0.00	0.00	0.00
3	N03	0.18	2.36	2.27	3.18	2.90	2.90
4	N04	0.00	0.00	0.00	0.00	2.85	0.00
5	N05	14.63	2.44	6.10	0.00	4.88	4.88
6	N06	7.74	0.00	0.00	7.74	0.00	0.00
7	N07	5.17	0.86	0.00	0.86	0.43	0.00
8	N09	2.70	2.70	0.00	0.00	0.00	0.00
9	N10	8.79	5.49	0.00	0.00	5.49	0.00
10	N11	21.14	1.63	1.63	0.00	1.63	0.00
11	N12	0.00	3.29	3.29	3.29	0.00	3.29
12	N13	13.26	0.00	15.47	0.00	0.00	0.00
13	N14	4.35	0.00	1.09	0.00	0.00	0.00
14	N15	0.00	0.00	0.00	0.00	0.00	0.00
15	N16	5.30	1.32	1.32	1.32	0.00	0.00
16	N17	4.85	0.00	0.00	0.00	0.00	4.85
17	N18	5.76	0.00	0.00	2.73	0.00	0.00
18	N21	0.00	5.33	0.00	5.33	0.00	5.33
19	N22	0.00	0.00	1.04	5.18	0.00	0.00
20	N23	0.00	0.00	0.00	0.00	0.00	0.00
21	N24	1.73	0.00	0.00	0.00	0.00	0.00
22	N25	4.94	0.00	0.00	0.00	0.00	4.94
23	N26	0.00	0.00	3.67	0.00	0.00	0.00
24	N27	15.92	8.66	0.00	8.66	0.00	0.00
25	N28	28.71	0.00	0.00	0.00	0.00	0.00
26	N29	0.00	0.00	0.00	0.00	0.00	0.00
27	N30	1.72	2.15	2.15	2.15	0.00	2.15
28	N31	0.00	0.00	0.00	0.00	0.00	0.00

No	Name	VAM	JHM	TOCM-MT	BCE	SSM	CSM
29	N32	0.00	0.00	0.00	0.00	2.25	0.00
30	N33	14.40	4.80	0.00	4.80	4.80	4.80
31	N34	0.00	0.00	0.00	0.00	0.00	0.00
32	N35	14.29	0.00	0.00	0.00	0.00	0.00
33	S1	0.00	10.98	0.00	0.95	10.98	0.00
34	S2	0.00	14.98	0.00	0.00	18.24	0.00
35	S3	0.00	0.00	0.00	0.00	0.00	0.00
36	S4	1.24	3.35	0.24	2.23	3.35	3.50
37	S5	0.00	0.00	0.00	0.00	0.00	0.00
38	Real 1	0.00	0.00	0.29	0.00	0.00	0.00
39	Real 2	0.00	0.00	0.32	0.00	0.00	0.00
40	Real 3	0.00	0.00	0.21	0.00	0.00	0.00
41	Real 4	0.00	0.00	0.35	0.00	0.00	0.00
42	Real 5	0.00	0.00	0.41	0.00	0.00	0.00
	AVERAGE	4.21	1.68	0.95	1.15	1.38	0.87

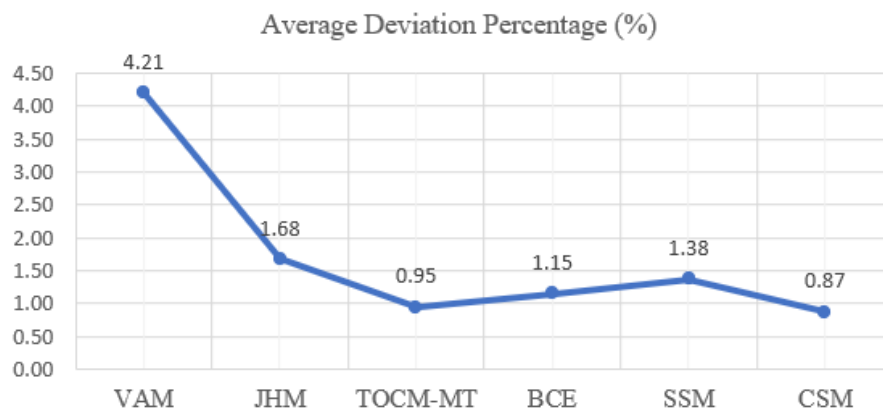


Figure 4.1. Average Deviation percentage among the methods.